

System Usability study (SUS)

After you implemented your system (Backend + frontend with proper GUI), you have to evaluate your system:

- Performance (accuracy / resource utilization) : from your side/ through the system with different testing tools.
- Usability test: with external users;

→ To measure the usability/ user friendliness/ HCI aspects/ UX of your application

→ A survey among actual user

The System Usability Scale (SUS) is a reliable tool for measuring the usability.

It consists of a 10 item questionnaire with five response options for respondents; from Strongly agree to Strongly disagree.

The participant's scores for each question are converted to a new number, added together and then multiplied by 2.5 to convert the original scores of 0-40 to 0-100.

Based on research, a SUS score above a 68 would be considered above average and anything below 68 is below average, however the best way to interpret your results involves "normalizing" the scores to produce a percentile ranking.

Statements of the SUS calculation process.

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

You can prepare a google form to collect data from the users.

Data collection Google form Example:

https://docs.google.com/forms/d/e/1FAIpQLScDfGMM2hxMS_Hq12PmnezLDWEtNzWJ6EsYqBL08tIqK5OzXA/viewform

- When you send the link of the system and link for the SUS google form, together with that send an overview document with a task list. (User Guide)

Prepare a user-guide with a task list/ sheet.

- That is a 1-2 page document, that tells users, what this system is for and the tasks they should do. (may be with some GUI)
- by having a formalized task list, the SUS can perform in a well-defined way.
- that is, all the users do the same set of tasks and the SUS results may become consistent; hence more accurate results.

In case, if you have implemented a login system

In the task sheet, it is better to indicate some user names and passwords (test accounts) that the user can use/ based on the role, only if needed for the system functions.

But make sure, not to give user accounts/ password/ access, in a way that user can harm to the system back end/ DB/ delete records/ etc.

In evaluation such as SUS, it is not good to ask to sign in from the beginning, by filling all the text boxes in the sign-in form. It consumes time and the task is obvious in any system. (in such an evaluation, the user tries to complete the task ASAP) . However, if a user prefers, he can create his own account and perform the task.

This Survey requires 20 – 25 users

After you entered the response number (in Likert scale) : 1 – 5 for each user for each question; it will generate the usability score and two charts.

Example write up

Usability Evaluation

The tool is evaluated using the System Usability Scale (SUS) with 20 users in the age group of 20-30 years, who have basic data visualization knowledge.

An average SUS score ofout of 100 was observed for the usability of the presented tool in this paper.

Explain the answers for the questions (example)

In the open-ended questionnaire, majority of the users indicated that the

Some users had stated that One major fact indicated by the users was that the

However, users gave positive reviews for the tool. Majority of the users indicated that it was simple and easy to use as all the steps were automated.

Example of n-gram (use a tool to draw it)

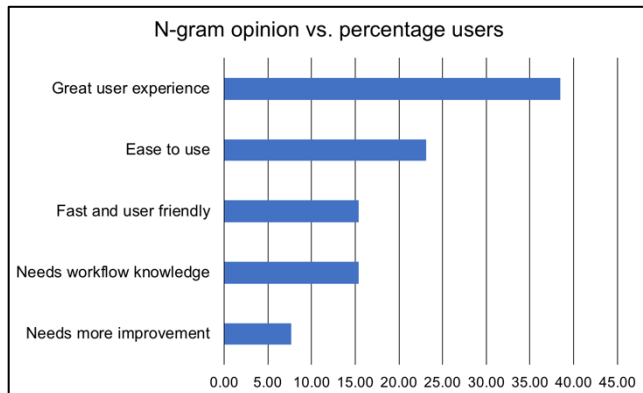


Figure. N-gram opinion analysis

Describe the n-gram (following is only an example. Use your own words and write again)

An n-gram opinion analysis was carried out on the feedback provided in the open-ended questionnaire regarding the tool. Figure .. illustrates the percentages of various opinions regarding the usability of the tool. More than 75% of the users were satisfied with the ease of use, simplicity and rapid generation of results. Only a few had moderate opinions and suggested minor improvements. On average, each of the users took about Minutes to experience the tool. Furthermore, some users had issues such as

References:

<https://measuringu.com/sus/>

J. Brooke, "SUS: a quick and dirty usability scale", In P.W.Jordan, B. Thomas, B.A. Weerdmeester, and I.L. McClelland (Eds.), Usability Evaluation in Industry, Taylor and Francis, pp. 189-194, 1996.

J. Brooke, "SUS: A retrospective", Journal of Usability Studies, vol. 8(2), 2013, pp. 29-40.